

TREMOR — Agentic Vibration Intelligence

Machine vibration measured from ordinary handheld video. OpenCV 5 recovers sub-pixel displacement; an agent decides what to measure next.

Team TREMOR · OpenCV AI Competition 2026 · Pursuing **both** featured paths: Best Use of COOL and Agentic Vision

Problem and intended real-world impact

Vibration analysis predicts rotating-machine failure before it happens: unbalance, misalignment and looseness each leave a distinct signature at multiples of shaft speed. It works, but it needs a contact accelerometer and a certified analyst, so most of the world's pumps, motors, fans and gearboxes are never measured — they run to failure. Camera-based motion amplification exists commercially (RDI, Erbesd, gfaitech), but every such product is a *visualisation tool*: it renders amplified video and hands it to the analyst, who remains the scarce, expensive part. **The gap is the analyst, not the algorithm.** TREMOR measures the vibration and judges whether it can trust the measurement — going back for a better clip when it cannot, and declining to answer when it still cannot.

Target users

Maintenance technicians and facilities staff at sites that cannot justify a vibration programme: mid-size plants, building services, campus and municipal estates, small manufacturers. People who own machines and own a phone.

OpenCV 5 image and video analysis

Every stage is a metric measurement on pixel data; no learned model appears anywhere in the measurement path. The output is a displacement of tens of micrometres at a named frequency — there is no prompt for that.

Stage	OpenCV 5 operation
1 Locate what vibrates	temporal high-pass per pixel, then cv2.boxFilter energy + texture maps
2 Sub-pixel displacement	cv2.phaseCorrelate with cv2.createHanningWindow
3 Reject failures	MAD plus phaseCorrelate's own correlation response
4 Cancel camera motion	static-reference subtraction, applied adaptively
5 Temporal spectrum	Hann window, cv2.dft, amplitude in pixels
6 Shaft rate and fault	harmonic-comb scoring, noise-floor-gated 1x/2x/3x ratios

AWS architecture and services

The hot path is classical imgproc/core — cv2.phaseCorrelate alone is 68% of measured operation time — which is exactly what COOL accelerates with Arm KleidiCV. There is no neural network in the measurement path, so the Graviton case is genuine rather than decorative. **Built and benchmarked:** the OpenCV 5 pipeline on EC2 c8g.2xlarge (Graviton4) under the COOL AML, and a FastAPI web endpoint. **Designed, not yet wired into the application:** S3/Lambda/SQS ingest and DynamoDB/CloudWatch state. Observability today is a per-step decision trace written to JSON and streamed live over SSE.

Measured results — this is a working system, not a plan

COOL vs stock OpenCV 5 on the *same* c8g.2xlarge instance, with a provenance gate proving the two legs loaded different libraries (COOL 5.1.0-dev from /opt/cool; baseline 5.0.0 from its own venv):

	ms/frame	x realtime	\$/video-hour	8-worker parallel efficiency
stock OpenCV 5.0.0	5.21	6.4	\$0.0498	94%
COOL 5.1.0-dev + KleidiCV	4.14	8.1	\$0.0419	92%
	1.26x		16% cheaper	vs 37% on Apple M-series

Real handheld iPhone clip, ground truth 7.30 Hz: **measured 7.276 Hz, 0.33% error**, while the operator's hand moved the camera 72 px peak-to-peak — 25x the signal being measured.

Agentic Vision: the visual evidence changes what happens next

The agent's tools are OpenCV operations, and every decision derives from a measurement rather than from prose. Guard rails live in the tools, not the prompt: diagnose() *refuses* when a needed harmonic exceeds Nyquist instead of returning a plausible wrong answer.

The measurement says	The agent does
texture below the correlation floor	re-aim at a grille, flange or label
SNR below 6, handheld	ask the operator to brace, then try a better aim point
a needed harmonic above Nyquist	re-acquire at 240 fps to resolve 3x
confidence below 0.55	escalate to a human with the evidence

The Nyquist case is the clearest: looseness is separable from unbalance only via its 3x harmonic, which sits above Nyquist at 30 fps for any shaft above 5 Hz. The agent derives this *from its own shaft estimate* and re-acquires. Measured effect: looseness went from 13 correct / 13 escalated to **19 correct / 7 escalated**.

Evaluation method and judge demonstration

Ground truth is exact by construction, so claims are testable rather than asserted. Agent task effectiveness at n=80: **71.2%** of scenarios diagnosed correctly, **87.7%** correct on those it committed to, 18.8% escalated rather than guessed, median shaft-frequency error **0.000 Hz**. Reported with a confusion matrix and Wilson 95% intervals. Rather than quote a single agent-vs-baseline ratio — which we found moved from 21% to 70% to 27% purely with scenario difficulty — we sweep that variable and report the curve: **agent accuracy is flat at 79.2% regardless of how badly the operator aims, while a single-shot baseline spans 8.3–75%**. The claim is invariance to a bad first acquisition, not a multiple.

Judges can run every figure from a clean checkout with no API key and no cloud account: `run_gate.py`, `run_machine_eval.py`, `eval_agent.py`, `sensitivity.py`, `pytest`. A live web endpoint streams the agent loop step by step so each decision appears with the evidence that drove it.

Limitations we state up front

Agent evaluation runs against a simulated acquisition environment we wrote, so those figures are evidence the loop behaves as designed, not evidence of field accuracy. Real-world validation is currently one clip of a controlled target. Frame rate caps the measurable band (30 fps → 15 Hz; 240 fps → 120 Hz), so bearing defect frequencies in the kHz are out of reach entirely.

Team

Team TREMOR — Tushar Agrawal, solo entrant. Software engineer working on production data and platform systems. Competition and hackathon experience; this is a first OpenCV Competition entry. Work to date is public at github.com/TusharTechs/tremorcv — measurement core, agent, MCP server, COOL/Graviton benchmark harness, web endpoint, technical report, and a threshold document recording the empirical basis for every constant including the results that went against us.

Featured paths pursued: **both** — Best Use of COOL (classical CV hot path on Graviton, benchmarked against a verified stock baseline on the same instance) and Agentic Vision (vision results choose the next acquisition).